Name: Piyush Misal .

PRN No:202501090107

Division:ME2

Batch:M21

Codetantra practical 3

3.1.1 problem statement

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

Code:

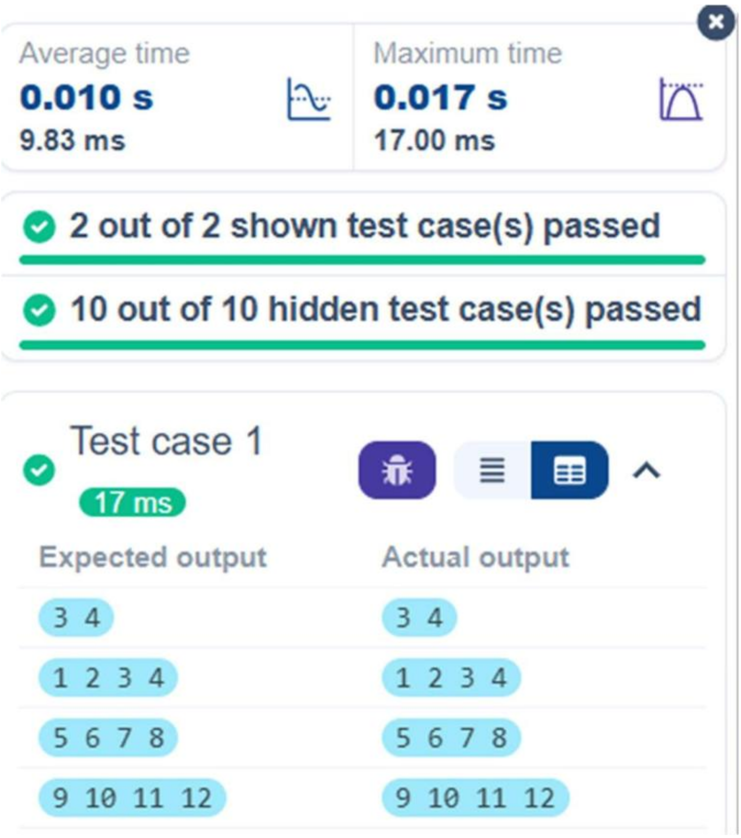
```
import numpy as np
r,c = map(int, input().split())
elements = []
for _ in range(r) :
    elements.extend(map(int, input().split()))
array = np.array(elements).reshape(r,c)

print(array)
```

print(array.ndim)

print(array.shape)

print(array.size)

Output:



[[·1··2··3··4]	[[·1··2··3··4]
·[·5··6··7··8]	·[·5··6··7··8]
·[·9·10·11·12]]	·[·9·10·11·12]]
2	2
(3,·4)	(3,·4)
12	12

Test case 2

3 ms

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0,·0)	(0,·0)
0	0

Practical Lab Assignment

3.2.1 problem statement

The given code takes two 3 * 3matrices,matrix_a , and matrix_b, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (matrix_a + matrix_b)

2. **Subtraction** (matrix_a – matrix_b)
3. **Element-wise Multiplication** (matrix_a * matrix_b)
4. **Matrix Multiplication** (matrix_a . matrix_b)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

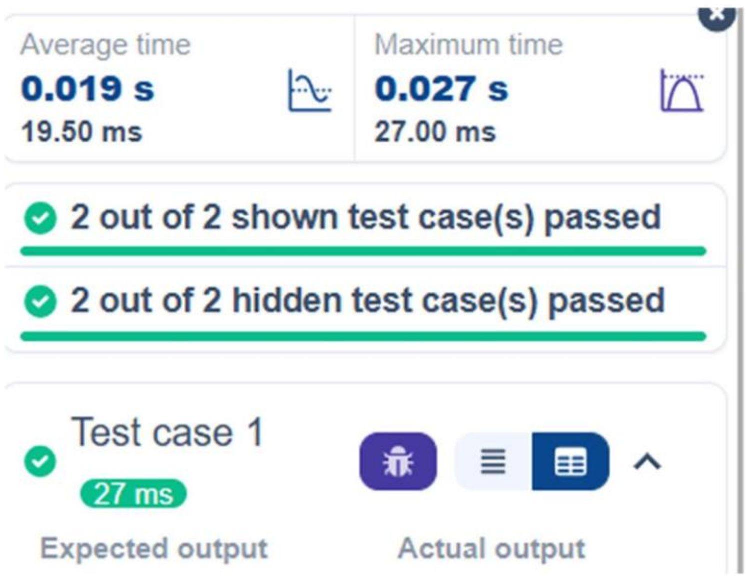
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np
```

```
# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i
in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i
in range(3)])



# Addition
addition_result=matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)
# Subtraction
subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)
# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a *
matrix_b
```

```
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)
# Matrix multiplication (dot product)
matrix_multiplication_result =np.dot(matrix_a,
matrix_b)
print("A dot B:")
print(matrix_multiplication_result)
# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```



Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]	[[2 3 4]
[5 6 7]	[5 6 7]
[8 9 10]]	[8 9 10]]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]	[[0 1 2]
[3 4 5]	[3 4 5]
[6 7 8]]	[6 7 8]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]]	[7 8 9]]
A dot B:	A dot B:
[[6 6 6]	[[6 6 6]
[15 15 15]	[15 15 15]

·[24·24·24]]⏎	·[24·24·24]]⏎
Transpose·of·A:⏎	Transpose·of·A:⏎
[[1·4·7]⏎	[[1·4·7]⏎
·[2·5·8]⏎	·[2·5·8]⏎
·[3·6·9]]⏎	·[3·6·9]]⏎

Test case 2

18 ms

Expected output	Actual output
Enter·Matrix·A:⏎	Enter·Matrix·A:⏎
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter·Matrix·B:⏎	Enter·Matrix·B:⏎
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition·(A·+·B):⏎	Addition·(A·+·B):⏎
[[12·16·21]⏎	[[12·16·21]⏎
·[23·21·21]⏎	·[23·21·21]⏎
·[24·26·28]]⏎	·[24·26·28]]⏎
Subtraction·(A·-·B):⏎	Subtraction·(A·-·B):⏎
[[·-8··-8··-5]⏎	[[·-8··-8··-5]⏎
·[·-5··-9·-11]⏎	·[·-5··-9·-11]⏎

[-10 -10 -10]]	[-10 -10 -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[20 48 104]	[[20 48 104]
[126 90 80]	[126 90 80]
[119 144 171]]	[119 144 171]]
A dot B:	A dot B:
[[212 228 242]	[[212 228 242]
[259 288 308]	[259 288 308]
[335 366 390]]	[335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]	[[2 9 7]
[4 6 8]	[4 6 8]
[8 5 9]]	[8 5 9]]

3.2.2 problem statement

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

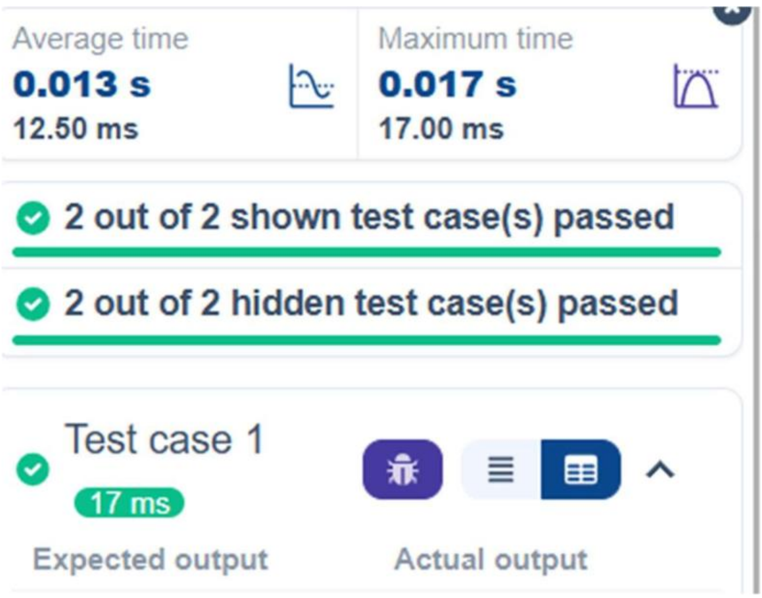
- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Array2:")
```

```
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
# Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
```

Output:



Enter·Array1:↵	Enter·Array1:↵
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter·Array2:↵	Enter·Array2:↵
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal·Stack:↵	Horizontal·Stack:↵
[[·1··2··3··4··5··6]↵	[[·1··2··3··4··5··6]↵
·[·4··5··6··7··8··9]↵	·[·4··5··6··7··8··9]↵
·[·7··8··9·10·11·12]]↵	·[·7··8··9·10·11·12]]↵
Vertical·Stack:↵	Vertical·Stack:↵
[[·1··2··3]↵	[[·1··2··3]↵
·[·4··5··6]↵	·[·4··5··6]↵
·[·7··8··9]↵	·[·7··8··9]↵
·[·4··5··6]↵	·[·4··5··6]↵
·[·7··8··9]↵	·[·7··8··9]↵
·[10·11·12]]↵	·[10·11·12]]↵

Test case 2

12 ms

Expected output	Actual output

Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[[-1 -2 -3 10 11 12]	[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]	[-7 -8 -9 16 17 18]]
Vertical Stack:	Vertical Stack:
[[-1 -2 -3]	[[-1 -2 -3]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]]	[16 17 18]]

3.2.3 problem statement

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.

- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

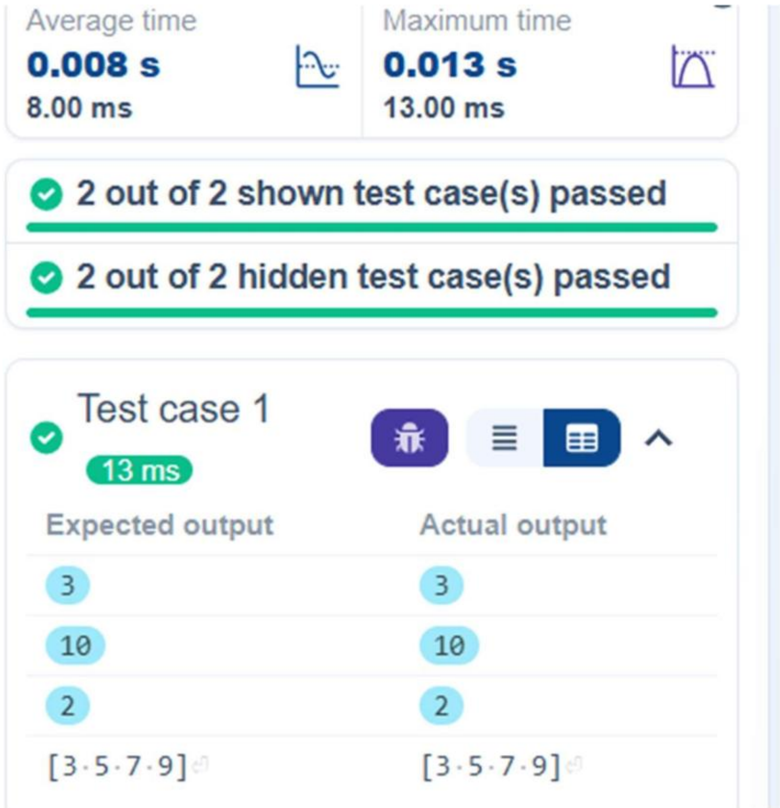
- The program should print the generated sequence based on the input values.

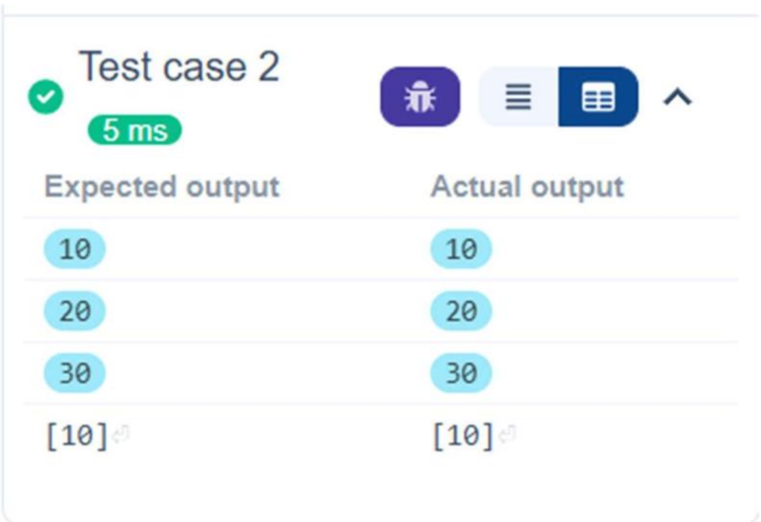
Code:

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
```

```
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
a=np.arange(start,stop,step)
print(a)
# Print the generated sequence
```

Output:





3.2.4 problem statement

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

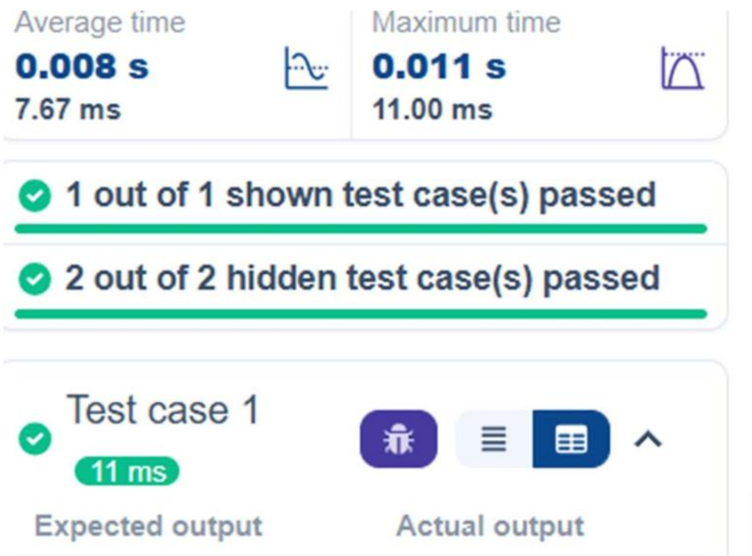
Code:

```
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)
    # Arithmetic Operations
    sum_result = A + B
```

```
diff_result = A - B
prod_result = A * B

# Statistical Operations
mean_A = np.mean(A)
median_A = np.median(A)
std_dev_A = np.std(A)

# Bitwise Operations
and_result = A & B
or_result =A | B
xor_result = A ^ B
  # Output results with one space between each
element
    print("Element-wise Sum:", ' '.join(map(str,
sum_result)))
    print("Element-wise Difference:", ' '.join(map(str,
diff_result)))
    print("Element-wise Product:", ' '.join(map(str,
prod_result)))
```

```
    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")
    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

Output:



1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5 problem statement

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

Code:

```
import numpy as np

inputlist = list(map(int,input().split(" ")))
```

```
# Original array
original_array = np.array(inputlist)

# Create a view
view_array =original_array.view()

# Create a copy
copy_array =original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)
# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

Output:

Average time

0.005 s

4.75 ms

Maximum time

0.010 s

10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

10 ms

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

Test case 2

3 ms

Expected output	Actual output
99 2 3 4 5	99 2 3 4 5
Original·array·after·modifying·view:·[99··2··3··4··5]	Original·array·after·modifying·view:·[99··2··3··4··5]
View·array:·[99··2··3··4··5]	View·array:·[99··2··3··4··5]
Original·array·after·modifying·copy:·[99··2··3··4··5]	Original·array·after·modifying·copy:·[99··2··3··4··5]
Copy·array:·[99·88··3··4··5]	Copy·array:·[99·88··3··4··5]

3.2.6 problem statement

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices
where search_value appears in array1 and print these indices.

2. Counting: Count how many
times count_value appears in array1 and print the count.

3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.

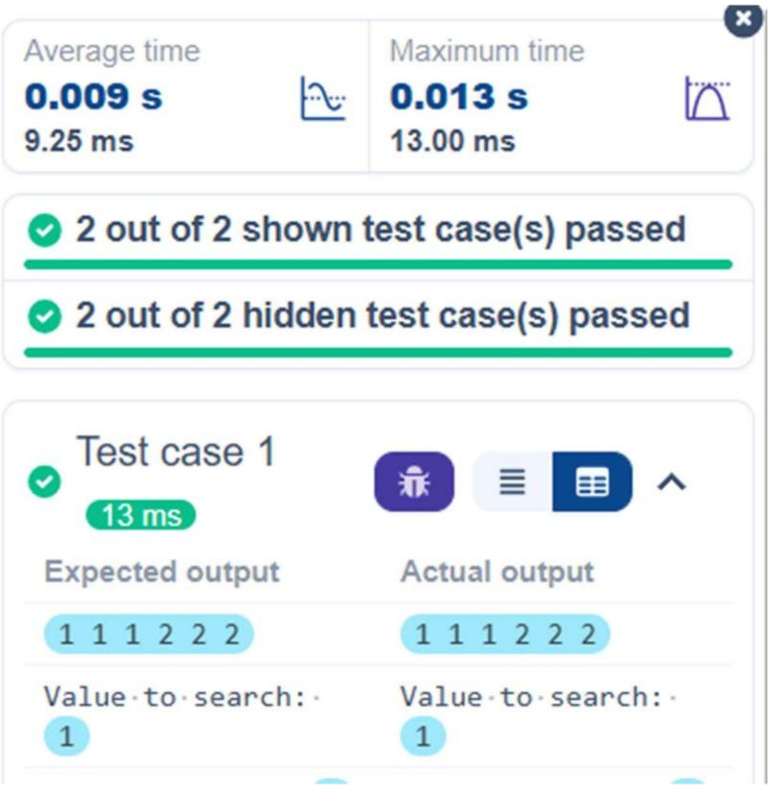
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np


# Input array from the user
array1 = np.array(list(map(int, input().split())))


# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
```

```
print(b)
# Broadcasting addition
c=array1 + broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
print(d)
```

Output:



Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3
[3 3 3 4 4 4]	[3 3 3 4 4 4]
[1 1 1 2 2 2]	[1 1 1 2 2 2]

Test case 2

7 ms

Expected output	Actual output
1 2 3 4 5	1 2 3 4 5
Value to search: 6	Value to search: 6
Value to count: 6	Value to count: 6
Value to add: 6	Value to add: 6
[]	[]
0	0
[7 8 9 10 11]	[7 8 9 10 11]
[1 2 3 4 5]	[1 2 3 4 5]

3.2.7 problem statement

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.

Find total marks of students: Compute the total marks for each student across all subjects..

- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.

- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.

- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in subject 1**:Identify the index positions of students who scored exactly 79 marks in subject 1.

Code:

```
import numpy as np


a = np.loadtxt("Sample.csv", delimiter=',',
skiprows=1)
# 1. Print all student details
print("All student Details:\n", a)
# 2. print total students
print("Total Students:", a.shape[0])
# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:, 0])
# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])
# 5. print minimum marks of Subject 2
```

```
print("Min marks in Subject 2", np.min(a[:, 2]))
# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))
# 7. Print All subject marks
print("All subject marks:", a[:, 1:])
total = np.sum(a[:, 1:], axis=1)
# 8. print Total marks of students
print("Total Marks", total)
#(no label in sample output)
print(np.round(np.mean(a[:, 1:], axis=1), 1))
# 9. print average marks of each student
# 10. print average marks of each subject
print("Average Marks of each subject",
np.round(np.mean(a[:, 1:], axis=0), 1))

# 11. print average marks of S1 and S2
print("Average Marks of S1 and S2",
np.round(np.mean(a[:, [1, 2]], axis=0), 1))
# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3",
np.round(np.mean(a[:, [1, 3]], axis=0), 1) )
```

```
# 13. print Roll number who got maximum marks in Subject 3

print("Roll no who got maximum marks in Subject 3", a[np.argmax(a[:, 3]), 0])

# 14. print Roll number who got minimum marks in Subject 2

print("Roll no who got minimum marks in Subject 2", a[np.argmin(a[:, 2]), 0])

# 15. print Roll number who got 24 marks in Subject 2

print("Roll no who got 24 marks in Subject 2", a[a[:, 2] == 24, 0].reshape(-1,1))

# 16. print count of students who got marks in Subject 1 < 40

print("Count of students who got marks in Subject 1 < 40", np.sum(a[:, 1] < 40))

# 17. print count of students who got marks in Subject 2 > 90

print("Count of students who got marks in Subject 2 > 90:", np.sum(a[:, 2] > 90))

# 18. print count of students in each subject who got marks >= 90
```

```
print("Count of students in each subject who got marks >= 90:",
        np.sum(a[:, 1:] >= 90, axis=0))
# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >= 90:",
np.sum(a[:, 1:] >= 90, axis=1))
# 20. Print S1 marks in ascending order
print(np.sort(a[:, 1]))
# 21. Print S1 marks >= 50 and <= 90
print(a[(a[:, 1] >= 50) & (a[:, 1] <= 90)])
print(a)
# 22. Print the index position of marks 79
print(np.where(a[:, 1] == 79))
```

Output:

Average time	Maximum time
0.014 s	**0.014 s**
14.00 ms	14.00 ms

1 out of 1 shown test case(s) passed

Test case 1

14 ms

Expected output	Actual output
All·student·Details:	All·student·Details:
·[[301.··67.··77.··88.]	·[[301.··67.··77.··88.]
·[302.··78.··88.··77.]	·[302.··78.··88.··77.]
·[303.··45.··56.··89.]	·[303.··45.··56.··89.]
·[304.··88.··98.··45.]	·[304.··88.··98.··45.]

[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]

[78. 88. 77.]	[78. 88. 77.]
[45. 56. 89.]	[45. 56. 89.]
[88. 98. 45.]	[88. 98. 45.]
[78. 88. 99.]	[78. 88. 99.]
[68. 78. 36.]	[68. 78. 36.]
[79. 12. 45.]	[79. 12. 45.]
[46. 45. 77.]	[46. 45. 77.]
[89. 32. 88.]	[89. 32. 88.]
[79. 24. 33.]]	[79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]	Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0

Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]	[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 7	[302. 78. 88.

9.]	99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
(array([6, 9], dtype=int32),)	(array([6, 9], dtype=int32),)